

PINGME

Advanced Ping Scanner v3.0

Complete Installation, Usage, Options, Workflows and Cheatsheet Manual

Subnet Analysis • ICMP/TCP Discovery • DNS • History • Diff • IPv4/IPv6

Tool	PingMe
Version	3.0
Author	Madhav
Document	Detailed User Manual

Authorized use only. Run PingMe only against systems and networks you own or are explicitly authorized to assess.

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Navigation tip. Use the PDF viewer search function for any flag, such as `--tcp-ports`, `--compare`, or `--resume`.

1. Introduction

PingMe is a zero-dependency Python network discovery utility designed for subnet analysis, host availability checks, IP classification, operational comparisons, and repeatable network assessment workflows. It combines ICMP probing with optional TCP connection checks, allowing it to identify systems that block ping but still expose approved services.

Core capabilities

- Analyze IPv4 and IPv6 networks without scanning.
- Discover hosts using `fping` or the operating system `ping` command.
- Treat a host as reachable when ICMP succeeds or a requested TCP port accepts a connection.
- Resolve hostnames and perform optional reverse DNS lookups.
- Estimate operating-system families from observed TTL values.
- Save scan history, compare repeated scans, resume interrupted scans, and diff snapshot files.
- Export reachable and non-responsive hosts as TXT, CSV, or JSON.

How PingMe decides that a host is reachable

A target is marked reachable when at least one ICMP reply is received, or when at least one port supplied with `--tcp-ports` accepts a TCP connection. This distinction is preserved in structured output using the `icmp_alive` and `tcp_open` fields.

Important. A host shown as non-responsive is not automatically powered off. Firewalls may block ICMP and all tested TCP ports. Validate important results with additional authorized tools.

2. Installation and Upgrade

2.1 Requirements

- Python 3.9 or newer.
- At least one ICMP backend: fping, ping, or ping6.
- No third-party Python packages are required.
- Administrator access is needed only for a system-wide Unix launcher under /usr/local/bin.

2.2 Recommended installer

```
git clone https://github.com/Madhav-Sai/pingme.git
cd pingme
python3 install.py
exec zsh
pingme --help
```

2.3 User-only installation

```
python3 install.py --user
exec zsh
pingme --help
```

The user installation normally places the launcher in ~/.local/bin. Ensure that directory is included in PATH.

2.4 Manual installation

```
chmod +x pingme.py
sudo ln -sfn "$(pwd)/pingme.py" /usr/local/bin/pingme
pingme --help
```

2.5 Optional fping installation

```
# Kali, Debian, Ubuntu
sudo apt update && sudo apt install -y fping

# Fedora/RHEL
sudo dnf install -y fping

# macOS
brew install fping
```

2.6 Shell completion

The installer can configure Zsh, Bash, Fish, or PowerShell completion. Completion suggests flags, help topics, output formats, ping backends, files, and directories.

```
pingme --<TAB>
pingme help <TAB>
pingme --out-format <TAB>
pingme --ping-tool <TAB>
```

2.7 Updating PingMe

```
cd ~/pingme
git pull
python3 install.py
exec zsh
```

3. Command Structure and Help

```
pingme [TARGET] [DISCOVERY] [SCAN CONTROL] [OUTPUT] [HISTORY]
```

3.1 Main help

```
pingme --help
```

3.2 Complete help

```
pingme --help-all
```

3.3 Nested help topics

```
pingme help targets
pingme help discovery
pingme help scan
pingme help output
pingme help history
pingme help advanced
pingme help examples
```

Compatibility. The nested help system improves discoverability without removing or changing the existing flags.

4. Target Selection

4.1 Analyze a subnet without scanning

```
pingme --sub 192.168.1.0/24
```

This prints the network address, broadcast address, subnet mask, wildcard mask, host range, usable count, scope, and possible subnet splits.

4.2 Scan a subnet

```
pingme --sub 192.168.1.0/24 --scan
```

4.3 Scan individual IPs and hostnames

```
pingme --host 10.0.0.10 server01.local 2001:db8::10
```

4.4 Scan a target file

```
# targets.txt
10.0.0.10
10.0.0.11
web01.example.local
2001:db8::25

pingme --file targets.txt
```

Blank lines, comments, duplicate addresses, and inline comments are handled. Unresolvable hostnames are skipped with a warning.

4.5 Exclusions

```
pingme --sub 10.0.0.0/24 --scan --exclude 10.0.0.1
pingme --sub 10.0.0.0/24 --scan --exclude 10.0.0.0/28 10.0.0.254
```

4.6 CIDR expansion safety

```
pingme --sub 10.10.0.0/16 --scan --max-hosts 65536
```

PingMe refuses subnet scans that exceed `--max-hosts`. This prevents accidental expansion of extremely large IPv4 or IPv6 networks.

5. Discovery Methods

5.1 ICMP discovery

By default PingMe uses auto backend selection. It prefers `fping` when available and otherwise falls back to the operating system ping implementation.

```
pingme --host 10.0.0.10 --ping-tool auto
pingme --host 10.0.0.10 --ping-tool fping
pingme --host 10.0.0.10 --ping-tool ping
```

5.2 TCP reachability checks

```
pingme --host 10.0.0.10 --tcp-ports 22,80,443
pingme --sub 10.0.0.0/24 --scan --tcp-ports 22,80,443,8000-8010
```

TCP checks are useful when a server or firewall drops ICMP. Port ranges are intentionally limited to prevent misuse and excessive accidental scanning.

5.3 Reverse DNS

```
pingme --file servers.txt --dns
```

5.4 TTL-based OS estimation

When ICMP replies expose a TTL, PingMe groups common values into broad families such as Windows, Linux/Unix, Cisco/Network, or Linux with many routing hops. This is an estimate, not definitive OS identification.

5.5 IP classification

```
pingme --ipinfo 8.8.8.8 192.168.1.1 100.64.0.1 127.0.0.1
```

Classification includes public, private, CGNAT, loopback, link-local, multicast, reserved, and documentation ranges where applicable.

6. Scan Control and Performance

6.1 Conservative default scan

```
pingme --sub 192.168.10.0/24 --scan
```

The defaults prioritize reliable detection: 20 workers, a six-second per-packet timeout, and eight packets per host.

6.2 Fast mode

```
pingme --sub 192.168.10.0/24 --scan --fast
```

Fast mode changes the scan to 100 workers, a one-second timeout, and one packet per host. Use it on low-latency networks when speed matters more than confidence.

6.3 Custom tuning

```
pingme --sub 10.0.0.0/22 --scan --threads 40 --timeout 3 --count 3 --retry 1
```

6.4 Rate limiting

```
pingme --sub 10.0.0.0/24 --scan --rate 100
```

The token-bucket limiter caps the approximate total packet rate across workers. Use it on fragile networks, remote links, VPNs, or during tightly controlled assessments.

6.5 TCP timeout

```
pingme --file servers.txt --tcp-ports 22,443 --tcp-timeout 3
```

6.6 Practical tuning guide

Option	Value	Purpose	Example
Reliable LAN	20 threads / 6s / 8 packets	Highest confidence for mixed devices	--threads 20 --timeout 6 --count 8
Fast LAN	100 / 1s / 1 packet	Quick local discovery	--fast
Remote VPN	10 / 5s / 4 packets	Lower load and tolerate latency	--threads 10 --timeout 5 --count 4
Fragile OT/IoT	5 / 6s / 3 packets + rate	Reduce pressure	--threads 5 --rate 20 --count 3
TCP-assisted	20 / defaults + ports	Find ICMP-blocked servers	--tcp-ports 22,80,443

7. Output and Reporting

7.1 TXT output

```
pingme --file targets.txt --alive-out alive.txt --dead-out dead.txt
```

TXT contains one IP per line and is ideal for piping into other authorized tools.

7.2 CSV output

```
pingme --file targets.txt --out-format csv --alive-out alive.csv --dead-out dead.csv
```

7.3 JSON output

```
pingme --file targets.txt --out-format json --alive-out alive.json --dead-out dead.json
```

7.4 Structured fields

Option	Value	Purpose	Example
ip	IP address	Resolved target address	10.0.0.10
alive	Boolean	ICMP or requested TCP	true

		check succeeded	
icmp_alive	Boolean	At least one ICMP response	false
tcp_open	List	Requested ports accepting connections	[22, 443]
ttl	Integer/null	Observed ICMP TTL	128
os_guess	String	Broad TTL-derived family	Windows
hostname	String	Reverse DNS result when --dns is used	server01.local
scope	String	Address classification	Private
rfc	String	Relevant RFC/authority	RFC 1918

7.5 Quiet and clean output

```
pingme --file targets.txt --quiet
pingme --file targets.txt --quiet --no-banner
```

8. History, Comparison and Resume

8.1 Automatic history

Unless --no-history is supplied, completed scans are appended to a JSON history file under ./data in the current working directory. A label groups repeated scans of the same environment.

8.2 Custom labels

```
pingme --sub 192.168.20.0/24 --scan --label office-lan
```

8.3 Compare with previous scan

```
pingme --sub 192.168.20.0/24 --scan --compare --label office-lan
```

8.4 List history

```
pingme --history
```

8.5 Clear history

```
pingme --clear-history office-lan
```

8.6 Disable history

```
pingme --file temporary_targets.txt --no-history
```

8.7 Resume interrupted scans

```
pingme --sub 10.0.0.0/16 --scan --label branch-network
# Press Ctrl+C if needed
pingme --sub 10.0.0.0/16 --scan --label branch-network --resume
```

8.8 Diff two snapshots

```
pingme --diff alive_monday.txt alive_friday.txt
```

9. Complete Option Reference

9.1 Targets

Option	Value	Purpose	Example
-s, --sub	CIDR	Analyze subnet; use --scan to probe hosts	--sub 192.168.1.0/24
-f, --file	FILE	Read IPs/hostnames from file; scan implied	--file targets.txt
--host	HOST ...	Resolve and scan one or more targets	--host web.local 10.0.0.5
--exclude	IP/CIDR ...	Remove addresses or networks	--exclude 10.0.0.1 10.0.0.0/28
--max-hosts	N	Maximum expanded CIDR targets	--max-hosts 4096

9.2 Discovery

Option	Value	Purpose	Example
--scan		Run discovery; required with --sub	--sub 10.0.0.0/24 --scan
--dns		Reverse DNS for reachable hosts	--dns
--tcp-ports	PORTS	TCP connect checks	--tcp-ports 22,80,443
--tcp-timeout	SEC	Timeout for each TCP connection	--tcp-timeout 3
--ipinfo	IP ...	Classify IP addresses	--ipinfo 8.8.8.8

9.3 Scan control

Option	Value	Purpose	Example
-t, --threads	N	Concurrent workers	--threads 40
--timeout	SEC	Per-packet wait time	--timeout 4
--count	N	Packets sent per host	--count 4
--retry	N	Additional attempts for dead hosts	--retry 1
--rate	PPS	Global packet rate cap; 0 unlimited	--rate 100
--ping-tool	auto fping ping	Choose backend	--ping-tool fping
--fast		100 workers, 1s, one packet	--fast
--resume		Continue partial scan for label	--resume

9.4 Output

Option	Value	Purpose	Example
--alive-out	FILE	Reachable target output	--alive-out up.json
--dead-out	FILE	Non-responsive target output	--dead-out down.json
--out-format	txt csv json	Output serialization	--out-format json
--label	NAME	History/resume grouping label	--label production
--quiet		Suppress individual host lines	--quiet

--no-banner		Disable ASCII banner	--no-banner
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9.5 History and help

Option	Value	Purpose	Example
--history		List history labels	--history
--compare		Compare current and previous label scans	--compare
--diff	A B	Compare two text snapshots	--diff old.txt new.txt
--clear-history	NAME	Delete a label history	--clear-history office
--no-history		Skip saving scan history	--no-history
--help-topic	TOPIC	Focused topic help	--help-topic scan
--help-all		Complete grouped help	--help-all
-h, --help		Main grouped help	--help

10. Real-World Workflows

Office morning availability check

```
pingme --sub 192.168.10.0/24 --scan --compare --label office-morning --dns
```

Shows newly online/offline systems and resolves names where possible.

Server inventory validation

```
pingme --file production_servers.txt --dns --tcp-ports 22,80,443,3389 --out-format csv --alive-out reachable.csv --dead-out unreachable.csv --label production
```

Combines ICMP and common approved service checks with a report-friendly CSV.

Quick authorized LAN discovery

```
pingme --sub 192.168.1.0/24 --scan --fast --no-history
```

Fast, disposable scan for a low-latency local segment.

Remote branch through VPN

```
pingme --sub 10.50.0.0/24 --scan --threads 10 --timeout 5 --count 4 --retry 1 --rate 50 --label branch-50
```

Reduces load and tolerates higher latency.

Find systems blocking ICMP

```
pingme --file servers.txt --tcp-ports 22,80,443,445,3389 --out-format json --alive-out reachable.json
```

A host can be reachable through TCP even if ping is blocked.

Exclude infrastructure devices

```
pingme --sub 10.20.30.0/24 --scan --exclude 10.20.30.1 10.20.30.2 10.20.30.240/28
```

Keeps gateways or out-of-scope ranges out of the assessment.

Weekly drift comparison

```
cp alive.txt alive_last_week.txt
pingme --sub 10.10.10.0/24 --scan
pingme --diff alive_last_week.txt alive.txt
```

Compares snapshot files without depending on stored history.

IPv6 small-range discovery

```
pingme --sub 2001:db8:1234::/120 --scan --max-hosts 256
```

Scans a deliberately bounded IPv6 range.

Pipeline only reachable IPs

```
pingme --file targets.txt --quiet --no-banner --alive-out alive.txt
cat alive.txt
```

Produces simple one-address-per-line output for later authorized processing.

11. Cheatsheets

11.1 Everyday commands

```
# Help
pingme --help
pingme help examples

# Subnet information only
pingme --sub 192.168.1.0/24

# Normal scan
pingme --sub 192.168.1.0/24 --scan

# Fast scan
pingme --sub 192.168.1.0/24 --scan --fast

# File scan with DNS and TCP
pingme --file targets.txt --dns --tcp-ports 22,80,443

# Compare repeat scan
pingme --sub 192.168.1.0/24 --scan --compare --label office

# Classify IPs
pingme --ipinfo 8.8.8.8 10.0.0.1 100.64.0.1

# Diff files
pingme --diff old_alive.txt new_alive.txt
```

11.2 Accuracy-first presets

```
# Mixed office devices
pingme --sub 192.168.10.0/24 --scan --threads 20 --timeout 6 --count 8

# Remote or lossy network
pingme --file targets.txt --threads 10 --timeout 6 --count 5 --retry 2

# Rate-controlled assessment
pingme --sub 10.0.0.0/24 --scan --threads 10 --rate 25
```

11.3 Output recipes

```
# Plain IP lists
pingme --file targets.txt --alive-out alive.txt --dead-out dead.txt

# CSV inventory
pingme --file targets.txt --dns --out-format csv --alive-out alive.csv --dead-out dead.csv

# JSON for scripts
pingme --file targets.txt --tcp-ports 22,443 --out-format json --alive-out alive.json
```

12. Troubleshooting

No ping tool found

Install fping or the OS ping package, then run pingme again. On Kali: `sudo apt install fping iputils-ping`.

fping requested but unavailable

Use `--ping-tool ping` or install fping.

Permission or raw socket error

Verify the system ping command works directly. Containers and restricted sandboxes may block ICMP sockets.

All hosts appear dead

Test a known live host, increase `--timeout` and `--count`, reduce `--threads`, and add approved `--tcp-ports`.

Host responds in browser but not PingMe

The host may block ICMP. Add the actual service port, for example `--tcp-ports 80,443`.

Scan is too slow

Use `--fast` on a reliable LAN, lower `--count`/`--timeout`, or increase `--threads` carefully.

Scan is too aggressive

Lower `--threads`, use `--rate`, and avoid broad TCP port lists.

Hostname skipped

Confirm DNS resolution with `getent hosts NAME`, `host NAME`, `nslookup NAME`, or `ping NAME`.

Tab completion not active

Re-run `python3 install.py` and reload the shell with `exec zsh`, `exec bash`, or restart the terminal.

History appears in unexpected folder

History is stored in `./data` relative to the directory from which PingMe is executed.

Resume does not find partial scan

Use the same `--label` and target selection that created the interrupted scan.

CSV/JSON output has wrong extension

PingMe writes the selected format to the exact filenames supplied; choose matching filename extensions yourself.

13. Accuracy, Limitations and Safety

- ICMP blocking can produce false-dead results unless relevant TCP checks are added.
- TCP connect checks only test the user-specified ports; closed or filtered ports do not prove the system is offline.
- TTL fingerprinting is approximate because routing hops, tunnels, NAT, and network devices modify the observed value.
- Reverse DNS depends on PTR records and resolver availability. A blank hostname is normal when no PTR record exists.
- Large numbers of workers or packets can stress constrained devices and links. Tune conservatively.
- IPv6 address spaces are enormous. Scan only explicitly bounded, authorized ranges.
- Output and history may contain sensitive internal addressing and hostnames. Store and share them appropriately.

Operational rule. Begin with a smaller target range and conservative settings. Confirm expected behavior before expanding scope.

14. Exit Behavior and Automation

PingMe is suitable for shell scripts because output files are deterministic and color is automatically disabled when stdout is not a terminal. Important argument and dependency failures exit non-zero.

14.1 Basic shell automation

```
#!/usr/bin/env bash
set -euo pipefail

pingme --file servers.txt --quiet --no-banner --no-history --tcp-ports 22,443 --out-format
json --alive-out reachable.json --dead-out unreachable.json

python3 -m json.tool reachable.json >/dev/null
echo "PingMe scan completed successfully"
```

14.2 Cron example

```
# Run at 08:00 every weekday from a fixed working directory
0 8 * * 1-5 cd /opt/network-checks && /usr/local/bin/pingme --file servers.txt --quiet --no-banner
--compare --label weekday-servers >> pingme.log 2>&1
```

Cron note. Use absolute paths and a fixed working directory so output and ./data history are stored predictably.

15. Quick Reference

Option	Value	Purpose	Example
Analyze subnet	pingme --sub CIDR	No traffic beyond local calculation	pingme --sub 10.0.0.0/24
Scan subnet	pingme --sub CIDR --scan	ICMP discovery	pingme --sub 10.0.0.0/24 --scan
Scan file	pingme --file FILE	IPs and hostnames	pingme --file targets.txt
TCP assist	--tcp-ports PORTS	Find ICMP-blocked services	--tcp-ports 22,80,443
DNS	--dns	PTR lookup	--dns
Fast	--fast	Speed over confidence	--fast
Compare	--compare --label NAME	Changes since prior scan	--compare --label office
Resume	--resume --label NAME	Continue interrupted scan	--resume --label office
JSON	--out-format json	Structured automation output	--out-format json
Help topic	pingme help TOPIC	Focused documentation	pingme help scan

Final checklist before a production scan

- Confirm written authorization and exact scope.
- Check exclusions and --max-hosts.
- Choose conservative threads, timeout, packet count, and rate.
- Select only necessary TCP ports.
- Use a meaningful --label for repeat scans.
- Choose output paths and protect resulting data.
- Validate results against at least one known-live and one known-offline target.